

LINEAR LEAST-SQUARES UNMIXING ON A SAMPLE NETWORK: DERIVATION, REGULARIZATION, AND ERROR PROPAGATION

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Alex G. Lipp¹

¹Department of Earth Sciences, University College London

ABSTRACT

Barnes and Lipp (2024) recover source concentrations on a river sample network by convex optimization, penalising *relative* differences between predicted and observed concentrations. Their Appendix A observes that penalising *absolute* differences instead reduces the problem to a linear least-squares matrix inversion. This note works that observation out in full. We show that the mixing matrix $\mathbf{M}$ is square, row-stochastic and triangular, hence always invertible, and we give its inverse in closed form as a local differencing stencil on the network. We then extend the formulation in three directions that the appendix does not cover: first-order tracer decay, weighting by the data covariance (generalised least squares), and Tikhonov regularization of the model variance. We prove that the regularized problem has a unique solution for every regularization strength, derive the resulting linear estimator, and obtain closed-form covariances for both the recovered sources and the modelled observations. We distinguish carefully between the propagated covariance and the Bayesian posterior covariance, which coincide only when the problem is unregularized, and give the exact identity relating them. Everything is derived from first principles so that it can be checked line by line.

1 SCOPE AND NOTATION

We adopt the notation of Barnes and Lipp (2024) throughout. The sample network is a directed acyclic graph $G = (N, E)$ whose n nodes are sample sites and whose edges point *downstream*. Each node i carries:

- a_i [L^2], the area of the sub-basin uniquely defined by sample site i ;
- ϕ_i [$ML^{-2}T^{-1}$], the material export rate of that sub-basin;
- $q_i = a_i\phi_i > 0$ [MT^{-1}], the material flux the sub-basin generates;
- c_i , the source concentration of the tracer in material exported by sub-basin i — the unknown;
- d_i , the concentration observed at the sample site — the datum.

Write $U_i \subseteq N$ for the set of nodes upstream of i *including i itself*, and $A(i)$ for the set of immediate upstream neighbours of i (its predecessors in G). We make one structural assumption throughout, which the D8 flow model guarantees:

(A1) Every node has at most one downstream neighbour, so G is a forest of in-trees.

Assumption (A1) is what makes the sets $\{U_p\}_{p \in A(i)}$ pairwise disjoint, which we use in Proposition 5. It does not require the network to be connected: disjoint river basins may be solved together, exactly as in the parent study.

Two derived quantities recur. The **total flux** passing site i is

$$Q_i = \sum_{j \in U_i} q_j, \quad (1)$$

and, for $j \in U_i$, the **path attenuation** from j to i is

$$\alpha_{j \rightarrow i} = \prod_{e \in \text{path}(j \rightarrow i)} \exp(-k_e L_e), \quad (2)$$

the product over edges of the unique path from j to i , where L_e is the flow-path length of edge e and k_e the first-order decay constant applied along it. By (A1) that path is unique, so (2) is well defined, and $\alpha_{i \rightarrow i} = 1$ (empty product). A **conservative** tracer has $k_e = 0$ for all e , hence $\alpha_{j \rightarrow i} = 1$ throughout; this is the case treated in Appendix A of the parent paper.

2 THE FORWARD MODEL AS A MATRIX

2.1 Assembling $\mathbf{M}$

Conservation of tracer at steady state gives the observation at site i as a flux-weighted mixture of every source upstream of it, attenuated by decay along the way:

$$d_i = \frac{\sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j}{\sum_{j \in U_i} q_j} = \frac{1}{Q_i} \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j. \quad (3)$$

Note that decay enters the numerator only. Tracer is lost in transit, but the *carrier* — the water or sediment — is not, so the denominator is the undecayed total flux Q_i . Setting every $\alpha = 1$ recovers Equation 3 of Barnes and Lipp (2024).

Following the parent appendix, define the 0–1 matrix Θ by $\Theta_{ij} = 1$ if and only if $j \in U_i$; that is, Θ is the path matrix of G with a self-loop added at every node. Equation (3) is then linear in $\mathbf{c}$, with

$$M_{ij} = \frac{\Theta_{ij} \alpha_{j \rightarrow i} q_j}{\sum_{k=1}^n \Theta_{ik} q_k} = \frac{\Theta_{ij} \alpha_{j \rightarrow i} q_j}{Q_i}, \quad \mathbf{d} = \mathbf{M}\mathbf{c}. \quad (4)$$

For a conservative tracer this is exactly Equation 11 of the parent appendix.

2.2 Structure of $\mathbf{M}$

$\mathbf{M}$ is $n \times n$: the sample network defines exactly one sub-basin per sample site, so there are as many unknowns as data. This squareness is the pivot on which everything below turns, and it is worth stating plainly that it is a modelling choice, not a coincidence — it is what it means to attribute one source concentration to each sampled sub-basin.

Lemma 1 (Row sums). *Every row of $\mathbf{M}$ is non-negative with $\sum_j M_{ij} \leq 1$, with equality for all i if and only if the tracer is conservative.*

Proof. Non-negativity is immediate from (4) since $q_j > 0$, $\alpha > 0$, $\Theta \in \{0, 1\}$. For the sum,

$$\sum_{j=1}^n M_{ij} = \frac{1}{Q_i} \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j \leq \frac{1}{Q_i} \sum_{j \in U_i} q_j = \frac{Q_i}{Q_i} = 1,$$

using $\alpha_{j \rightarrow i} \leq 1$, which holds because every $k_e L_e \geq 0$. Equality for a given i requires $\alpha_{j \rightarrow i} = 1$ for all $j \in U_i$, i.e. no decay anywhere upstream of i ; equality for all i requires it everywhere. $\square$

For a conservative tracer, then, $\mathbf{M}$ is **row-stochastic**: each observation is a convex combination of the sources upstream of it. This is the formal statement of “a river sample is a mixture of its upstream sources”, and it has a useful consequence, $\mathbf{M}\mathbf{1} = \mathbf{1}$, which we use in Section 7.1.

Lemma 2 (Triangularity). *Let σ be any topological ordering of G , so that $\sigma(u) < \sigma(v)$ whenever $(u, v) \in E$. Permuting rows and columns of $\mathbf{M}$ by σ makes it lower triangular, with diagonal entries*

$$M_{ii} = \frac{q_i}{Q_i} \in (0, 1]. \quad (5)$$

Proof. $M_{ij} \neq 0$ requires $\Theta_{ij} = 1$, i.e. $j \in U_i$. If $j \neq i$ then a directed path runs from j to i , so $\sigma(j) < \sigma(i)$. Hence $M_{ij} \neq 0 \Rightarrow \sigma(j) < \sigma(i)$, which is lower triangularity in the permuted ordering. The diagonal follows from (4) with $j = i$, using $\Theta_{ii} = 1$ and $\alpha_{i \rightarrow i} = 1$; it is positive because $q_i > 0$, and at most 1 because $q_i \leq Q_i$, with $M_{ii} = 1$ exactly when i is a leaf (headwater) node. $\square$

Proposition 3 (Invertibility). *$\mathbf{M}$ is invertible, with*

$$|\det \mathbf{M}| = \prod_{i=1}^n \frac{q_i}{Q_i} > 0. \quad (6)$$

Proof. Let Π be the permutation matrix of the topological order σ . By Lemma 2, $\Pi \mathbf{M} \Pi^T$ is lower triangular, so its determinant is the product of its diagonal entries, $\prod_i q_i / Q_i$, which is strictly positive because every $q_i > 0$ and every $Q_i < \infty$. Since $|\det \Pi| = 1$, $|\det \mathbf{M}| = \prod_i q_i / Q_i > 0$, and a matrix with non-zero determinant is invertible. $\square$

Remark 4. Invertibility is unaffected by decay: α appears only off the diagonal, so (5) is untouched. It is also unaffected by the export rates ϕ , provided they are strictly positive. It *would* fail if some sub-basin had zero area or zero export rate, i.e. if two sample sites were co-located — then $q_i = 0$, the corresponding source is not identifiable, and the model is degenerate. This is worth checking in practice.

3 THE INVERSE IN CLOSED FORM

Proposition 3 says $\mathbf{c} = \mathbf{M}^{-1}\mathbf{d}$ exists and is unique. We now show it need never be computed by matrix inversion: it is a local differencing stencil on the network, computable in one sweep.

It is convenient to work with the **tracer flux** passing site i ,

$$T_i = Q_i d_i = \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j, \quad (7)$$

which is just (3) cleared of its denominator. T_i has units of tracer mass per time: it is the amount of tracer passing the sample site per unit time.

Proposition 5 (Closed-form inverse). *For every node i ,*

$$c_i = \frac{1}{q_i} \left(Q_i d_i - \sum_{p \in A(i)} \alpha_{p \rightarrow i} Q_p d_p \right). \quad (8)$$

Proof. Every node in U_i other than i lies upstream of exactly one immediate neighbour $p \in A(i)$, and by (A1) the sets U_p for distinct $p \in A(i)$ are pairwise disjoint. Hence

$$U_i = \{i\} \sqcup \bigsqcup_{p \in A(i)} U_p,$$

a disjoint union. Now fix $p \in A(i)$ and $j \in U_p$. The unique path $j \rightarrow i$ is the path $j \rightarrow p$ followed by the single edge $p \rightarrow i$, so the attenuation factorises:

$$\alpha_{j \rightarrow i} = \alpha_{j \rightarrow p} \alpha_{p \rightarrow i}. \quad (9)$$

Substituting the partition and (9) into (7), and using $\alpha_{i \rightarrow i} = 1$ for the $j = i$ term,

$$\begin{aligned} T_i &= q_i c_i + \sum_{p \in A(i)} \sum_{j \in U_p} \alpha_{j \rightarrow i} q_j c_j \\ &= q_i c_i + \sum_{p \in A(i)} \alpha_{p \rightarrow i} \underbrace{\sum_{j \in U_p} \alpha_{j \rightarrow p} q_j c_j}_{= T_p} \\ &= q_i c_i + \sum_{p \in A(i)} \alpha_{p \rightarrow i} T_p. \end{aligned} \quad (10)$$

Equation (10) is a statement of tracer conservation at node i : the tracer leaving equals the tracer generated locally plus the (attenuated) tracer delivered by the immediate tributaries. Solving for c_i and substituting $T = Qd$ gives (8). Since $q_i > 0$ the division is legitimate. $\square$

Corollary 6 (Sparsity of $\mathbf{M}^{-1}$).

$$(\mathbf{M}^{-1})_{ij} = \begin{cases} \frac{Q_i}{q_i}, & j = i, \\ -\frac{\alpha_{j \rightarrow i} Q_j}{q_i}, & j \in A(i), \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

$\mathbf{M}^{-1}$ therefore has at most $n + |E| \leq 2n - 1$ non-zero entries, against up to $n(n + 1)/2$ in $\mathbf{M}$ itself. Equation (8) evaluates in $O(n + |E|)$ time.

It is worth pausing on how much structure this reveals. $\mathbf{M}$ is dense and global — every downstream observation depends on every source above it — yet its inverse is sparse and purely local. Recovering a source concentration requires only the observation at that site and the observations at its immediate tributaries. All the long-range mixing cancels.

Corollary 7 (Worked example). *For the six-node network of Barnes and Lipp (2024, Figure 2), with $A(4) = \{1, 2\}$, $A(5) = \{3\}$, $A(6) = \{4, 5\}$ and no decay, Equation (8) gives*

$$\begin{aligned} c_1 &= d_1, & c_2 &= d_2, & c_3 &= d_3, \\ c_4 &= \frac{Q_4 d_4 - Q_1 d_1 - Q_2 d_2}{q_4}, \\ c_5 &= \frac{Q_5 d_5 - Q_3 d_3}{q_5}, \\ c_6 &= \frac{Q_6 d_6 - Q_4 d_4 - Q_5 d_5}{q_6}, \end{aligned}$$

with $Q_1 = q_1$, $Q_2 = q_2$, $Q_3 = q_3$, $Q_4 = q_1 + q_2 + q_4$, $Q_5 = q_3 + q_5$ and $Q_6 = \sum_{i=1}^6 q_i$. Headwater sites are recovered exactly and trivially; interior sites are recovered by differencing.

3.1 Noise amplification

Differencing is the source of all the difficulty. Suppose the observations carry independent errors of standard deviation σ_i . Propagating them through (8) (anticipating Section 8, and using independence),

$$\text{Var}(c_i) = \frac{Q_i^2 \sigma_i^2 + \sum_{p \in A(i)} \alpha_{p \rightarrow i}^2 Q_p^2 \sigma_p^2}{q_i^2}. \quad (12)$$

Every term is scaled by q_i^{-2} . Define the **amplification factor**

$$\kappa_i = \frac{Q_i}{q_i} = \frac{1}{M_{ii}} \geq 1, \quad (13)$$

the ratio of the total flux passing site i to the flux its own sub-basin generates. If site i sits immediately downstream of another site, with little incremental drainage area between them, then $q_i \ll Q_i$, κ_i is large, and (8) subtracts two nearly equal large numbers to obtain a small one. This is catastrophic cancellation, and it is an intrinsic property of the sampling design, not of the algorithm: the data simply do not constrain that sub-basin well. In practice κ_i , being free to compute, is the single most useful diagnostic of whether a linear inversion is worth attempting on a given network.

Corollary 8 (When positivity binds). *Since concentrations must be non-negative, the unregularized solution is physically admissible at node i if and only if*

$$Q_i d_i \geq \sum_{p \in A(i)} \alpha_{p \rightarrow i} Q_p d_p, \quad (14)$$

that is, if and only if at least as much tracer flux leaves site i as its tributaries deliver to it. A violation is a direct falsification of the forward model (3) by the data, whether through measurement error, non-conservative behaviour, or mis-specified fluxes.

4 WHY THE UNREGULARIZED PROBLEM IS DEGENERATE

Appendix A of Barnes and Lipp (2024) poses the recovery as a constrained least-squares problem,

$$\underset{\mathbf{c} \in \mathbb{R}^n}{\text{minimize}} \quad \|\mathbf{M}\mathbf{c} - \mathbf{d}\|_2 \quad \text{subject to} \quad c_i \geq 0. \quad (15)$$

Proposition 3 shows that, absent the constraint, this is not really a least-squares problem at all.

Proposition 9 (Degeneracy). *Let $\mathbf{W}$ be any symmetric positive definite weight matrix. Then*

$$\arg \min_{\mathbf{c} \in \mathbb{R}^n} (\mathbf{M}\mathbf{c} - \mathbf{d})^\top \mathbf{W} (\mathbf{M}\mathbf{c} - \mathbf{d}) = \mathbf{M}^{-1} \mathbf{d}, \quad (16)$$

independently of $\mathbf{W}$, and the residual at the optimum is exactly zero.

Proof. The objective is non-negative for every $\mathbf{c}$, and $\mathbf{c} = \mathbf{M}^{-1} \mathbf{d}$ — which exists by Proposition 3 — attains the value 0. Hence it is a global minimiser, and it is the unique one because $\mathbf{W} > 0$ forces the residual to vanish, and $\mathbf{M}$ is injective. Algebraically, the generalised least-squares estimator collapses:

$$(\mathbf{M}^\top \mathbf{W} \mathbf{M})^{-1} \mathbf{M}^\top \mathbf{W} \mathbf{d} = \mathbf{M}^{-1} \mathbf{W}^{-1} \mathbf{M}^{-\top} \mathbf{M}^\top \mathbf{W} \mathbf{d} = \mathbf{M}^{-1} \mathbf{d}. \quad \square$$

Three consequences follow, and they set up the rest of this note.

1. **Weighting is inert when $\lambda = 0$.** No choice of $\mathbf{W}$ — including $\mathbf{W} = \mathbf{C}_d^{-1}$, the statistically correct one — changes the answer. There is nothing to trade off when the model can fit every datum exactly. Weighting only becomes consequential once a competing term is added to the objective (Section 7).
2. **The problem interpolates the noise.** A zero residual means the estimate reproduces the observations exactly, errors included. Combined with the amplification of Section 3.1, this is why the raw inversion is unusable on realistic data.
3. **The constraint is the only thing that can bite.** By Corollary 8, (15) departs from $\mathbf{M}^{-1}\mathbf{d}$ precisely when (14) fails somewhere.

Remark 10 (Bounds). The parent appendix writes the constraint as $0 < c_i < 1$, appropriate for a tracer measured as a mass fraction. Two remarks. First, a strict inequality cannot be imposed in a convex program — the feasible set would not be closed and the infimum need not be attained — so one uses $c_i \geq 0$. Second, the upper bound is unit-dependent (1 for a fraction, 10^6 for mg kg^{-1}) whereas the lower bound is not. Since $\mathbf{M}$ is row-substochastic, an estimate exceeding the physical ceiling indicates that the model has already failed, and is more useful surfaced as a diagnostic than silently clipped. Below we impose $c_i \geq 0$ only.

5 SCALE INVARIANCE AND NORMALISATION

Proposition 11 (Homogeneity). *The unregularized solution map $\mathbf{d} \mapsto \mathbf{c}$ is homogeneous of degree one: $\mathbf{c}(t\mathbf{d}) = t\mathbf{c}(\mathbf{d})$ for any $t > 0$. Moreover $\mathbf{M}$ is invariant under a common rescaling of the fluxes, $q_j \mapsto sq_j$.*

Proof. The first claim is immediate from linearity, $\mathbf{c} = \mathbf{M}^{-1}\mathbf{d}$. For the second, replacing q_j by sq_j in (4) multiplies numerator and denominator by s . $\square$

Homogeneity licenses the mean normalisation used in the implementation. Writing $\bar{d} = n^{-1} \sum_i d_i$ and solving with $\tilde{\mathbf{d}} = \mathbf{d}/\bar{d}$, the recovered sources are $\mathbf{c} = \bar{d}\tilde{\mathbf{c}}$ exactly. This changes nothing mathematically; its purpose is numerical, keeping all quantities near unity so that solver tolerances — which are partly absolute — behave uniformly across tracers whose concentrations may differ by many orders of magnitude. It has the further benefit of rendering the regularization strength introduced below dimensionless, and hence transferable between tracers and datasets.

An identical argument applies to the sub-basin areas, which the implementation divides by their mean, and to covariances, which by Proposition 11 and Section 8 scale as $\bar{d}^2$ and so may equivalently be computed in the original units.

6 WEIGHTING BY THE DATA COVARIANCE

Let $\mathbf{C}_d > 0$ be the covariance of the observation errors. The statistically natural misfit is the Mahalanobis norm of the residual, giving generalised least squares (Aster et al. 2018; Menke 2012):

$$\Phi(\mathbf{c}) = (\mathbf{M}\mathbf{c} - \mathbf{d})^T \mathbf{C}_d^{-1} (\mathbf{M}\mathbf{c} - \mathbf{d}). \quad (17)$$

Numerically one never forms $\mathbf{C}_d^{-1}$. Take the Cholesky factorisation $\mathbf{C}_d = \mathbf{L}\mathbf{L}^T$ (Golub and Van Loan 2013) and define the whitened system

$$\tilde{\mathbf{M}} = \mathbf{L}^{-1}\mathbf{M}, \quad \tilde{\mathbf{d}} = \mathbf{L}^{-1}\mathbf{d}, \quad (18)$$

so that $\Phi(\mathbf{c}) = \|\tilde{\mathbf{M}}\mathbf{c} - \tilde{\mathbf{d}}\|_2^2$: generalised least squares becomes ordinary least squares on the whitened system, whose residuals have identity covariance. When $\mathbf{C}_d$ is diagonal this reduces to dividing row i of $\mathbf{M}$ and d_i by σ_i .

Remark 12 (Relation to the log-ratio objective). Suppose errors are proportional, $\sigma_i = fd_i$ for a constant relative error f — the standard model for geochemical assay. The i th whitened residual is then

$$\frac{(\mathbf{M}\mathbf{c} - \mathbf{d})_i}{fd_i} = \frac{1}{f} \left(\frac{(\mathbf{M}\mathbf{c})_i}{d_i} - 1 \right), \quad (19)$$

which depends on the data only through the *ratio* of predicted to observed concentration. The weighted linear objective is therefore, like the log-ratio objective of the parent study, a relative misfit; it is the unweighted objective that is not. This is precisely the objection Barnes and Lipp (2024) raise against Euclidean misfit — that it over-weights high concentrations — and weighting by a proportional error model answers it. The two remain distinct: (17) penalises the squared relative residual, whereas $\max(x, 1/x)$ with $x = d_{\text{pred}}/d_{\text{obs}}$ behaves like $1 + |x - 1|$ near $x = 1$. Both are functions of the relative residual alone, but one is quadratic and the other piecewise linear in it, so they weight outliers differently and do not coincide.

7 REGULARIZATION

Section 4 showed that the unregularized problem interpolates the noise. The remedy is Tikhonov regularization (Hansen 1998; Hoerl and Kennard 1970; Tikhonov 1963): add a penalty expressing a preference among models.

We penalise the **variance of the model about its own mean**. Let

$$\mathbf{P} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \quad (20)$$

be the centering projector, so that $(\mathbf{P}\mathbf{c})_i = c_i - \bar{c}$. The full problem is

$$\begin{array}{ll} \underset{\mathbf{c} \in \mathbb{R}^n}{\text{minimize}} & J(\mathbf{c}) = (\mathbf{M}\mathbf{c} - \mathbf{d})^\top \mathbf{C}_d^{-1}(\mathbf{M}\mathbf{c} - \mathbf{d}) + \lambda \|\mathbf{P}\mathbf{c}\|_2^2 \\ \text{subject to} & \mathbf{c} \geq 0. \end{array} \quad (21)$$

Note that this penalises departure from the mean of the *model*, not, as in the parent study, from the mean of the *observations*. That substitution was forced there by the requirement that the objective be expressible in disciplined convex form (Boyd and Vandenberghe 2004; Diamond and Boyd 2016); the linear formulation is under no such constraint and can use the more natural quantity.

Lemma 13 (Properties of $\mathbf{P}$). *$\mathbf{P}$ is symmetric and idempotent, hence an orthogonal projector; it is positive semi-definite with eigenvalue 0 on $\text{span}\{\mathbf{1}\}$ and eigenvalue 1 on its orthogonal complement; and*

$$\|\mathbf{P}\mathbf{c}\|_2^2 = \mathbf{c}^\top \mathbf{P}^\top \mathbf{P} \mathbf{c} = \mathbf{c}^\top \mathbf{P} \mathbf{c} = \sum_{i=1}^n (c_i - \bar{c})^2. \quad (22)$$

Proof. Symmetry is clear. For idempotency, using $\mathbf{1}^\top \mathbf{1} = n$,

$$\mathbf{P}^2 = \mathbf{I} - \frac{2}{n}\mathbf{1}\mathbf{1}^\top + \frac{1}{n^2}\mathbf{1}(\mathbf{1}^\top \mathbf{1})\mathbf{1}^\top = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top = \mathbf{P}.$$

Then $\mathbf{P}^\top \mathbf{P} = \mathbf{P}^2 = \mathbf{P}$, giving the quadratic form. $\mathbf{P}\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0}$, and for $\mathbf{v} \perp \mathbf{1}$, $\mathbf{P}\mathbf{v} = \mathbf{v}$. Eigenvalues in $\{0, 1\}$ give $\mathbf{P} \geq 0$. $\square$

$\mathbf{P}$ is thus singular, with a one-dimensional null space. This matters: the penalty alone does not determine a unique solution, and the classical uniqueness condition for general-form Tikhonov regularization (Eldén 1977; Hansen 1998) requires $\mathcal{N}(\mathbf{M}) \cap \mathcal{N}(\mathbf{P}) = \{\mathbf{0}\}$. Here $\mathcal{N}(\mathbf{M}) = \{\mathbf{0}\}$ by Proposition 3, so the condition holds trivially — but it holds *because of* the structure of $\mathbf{M}$, not because of the penalty.

Proposition 14 (Existence and uniqueness). *For every $\lambda \geq 0$, J is strictly convex on $\mathbb{R}^n$. Consequently (21) has a unique global minimiser, both unconstrained and subject to $\mathbf{c} \geq 0$.*

Proof. J is a sum of quadratic forms, with gradient and Hessian

$$\nabla J(\mathbf{c}) = 2\mathbf{M}^\top \mathbf{C}_d^{-1}(\mathbf{M}\mathbf{c} - \mathbf{d}) + 2\lambda \mathbf{P}\mathbf{c}, \quad (23)$$

$$\nabla^2 J = 2(\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} + \lambda \mathbf{P}). \quad (24)$$

For any $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^\top \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} \mathbf{v} = (\mathbf{M}\mathbf{v})^\top \mathbf{C}_d^{-1}(\mathbf{M}\mathbf{v}) > 0$, since $\mathbf{M}\mathbf{v} \neq \mathbf{0}$ by Proposition 3 and $\mathbf{C}_d^{-1} > 0$. Hence $\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} > 0$. By Lemma 13, $\lambda \mathbf{P} \geq 0$ for $\lambda \geq 0$. A positive definite matrix plus a positive semi-definite one is positive definite (Horn and Johnson 2012), so $\nabla^2 J > 0$ and J is strictly convex. A strictly convex function has at most one minimiser on any convex set, and coercivity (guaranteed by $\nabla^2 J > 0$) ensures one exists. The non-negative orthant is convex and closed, so the constrained problem inherits both. $\square$

Remark 15. Uniqueness rests on $\mathbf{M}$ being invertible, *not* on the penalty. Had $\mathbf{M}$ been singular with a constant vector in its null space, $\lambda \mathbf{P}$ could not have restored uniqueness, because $\mathbf{P}$ is blind to exactly that direction. Regularization here buys conditioning and stability, not identifiability — the problem was already identifiable.

Proposition 16 (The estimator). *Define*

$$\mathbf{A}_\lambda = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} + \lambda \mathbf{P}. \quad (25)$$

If the unconstrained minimiser of J is non-negative, it is also the solution of (21), and it is given by the linear map

$$\hat{\mathbf{c}} = \mathbf{R}_\lambda \mathbf{d}, \quad \mathbf{R}_\lambda = \mathbf{A}_\lambda^{-1} \mathbf{M}^\top \mathbf{C}_d^{-1}. \quad (26)$$

Proof. Setting (23) to zero gives the normal equations $(\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} + \lambda \mathbf{P})\mathbf{c} = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{d}$, i.e. $\mathbf{A}_\lambda \mathbf{c} = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{d}$. $\mathbf{A}_\lambda > 0$ by the proof of Proposition 14, hence invertible, giving (26). For the first claim: J is strictly convex, so $\hat{\mathbf{c}}$ is its unique global minimiser over all of $\mathbb{R}^n$; if it happens to be feasible then it is a fortiori the minimiser over the smaller feasible set. $\square$

Two features of (26) deserve emphasis. First, it is exactly linear in $\mathbf{d}$, with **no constant offset**. This is a consequence of penalising $\|\mathbf{P}\mathbf{c}\|^2$, which is homogeneous in $\mathbf{c}$; a penalty $\|\mathbf{c} - \mathbf{c}_0\|^2$ towards a fixed reference $\mathbf{c}_0 \neq \mathbf{0}$ would contribute an affine term and complicate everything downstream. Second, Proposition 16 gives a practical recipe: solve the problem, and if the answer is non-negative, report the analytical $\mathbf{R}_\lambda \mathbf{d}$. The estimate and its covariance then provably come from the same linear operator, which is what makes the error propagation of Section 8 meaningful. When the constraint is active the estimator is only piecewise linear, and that guarantee is lost.

7.1 Limits

Proposition 17 (Small λ). $\mathbf{R}_0 = \mathbf{M}^{-1}$, recovering Section 3.

Proof. Set $\lambda = 0$ in (26): $\mathbf{R}_0 = (\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M})^{-1} \mathbf{M}^\top \mathbf{C}_d^{-1} = \mathbf{M}^{-1}$, as in Proposition 9. $\square$

Proposition 18 (Large λ). As $\lambda \rightarrow \infty$, $\hat{\mathbf{c}} \rightarrow t^* \mathbf{1}$ with

$$t^* = \frac{\mathbf{m}^\top \mathbf{C}_d^{-1} \mathbf{d}}{\mathbf{m}^\top \mathbf{C}_d^{-1} \mathbf{m}}, \quad \mathbf{m} = \mathbf{M} \mathbf{1}. \quad (27)$$

If in addition the tracer is conservative and $\mathbf{C}_d = \sigma^2 \mathbf{I}$, then $t^* = \bar{d}$, the arithmetic mean of the observations.

Proof. The penalty term $\lambda \|\mathbf{P}\mathbf{c}\|^2$ diverges unless $\mathbf{P}\mathbf{c} \rightarrow \mathbf{0}$, i.e. unless $\mathbf{c}$ approaches $\mathcal{N}(\mathbf{P}) = \text{span}\{\mathbf{1}\}$. Writing $\mathbf{c} = t\mathbf{1}$ the penalty vanishes identically and the problem reduces to minimising $(t\mathbf{m} - \mathbf{d})^\top \mathbf{C}_d^{-1} (t\mathbf{m} - \mathbf{d})$ over the scalar t . Differentiating and setting to zero gives t^* as stated. For a conservative tracer Lemma 1 gives $\mathbf{m} = \mathbf{M} \mathbf{1} = \mathbf{1}$, whence with $\mathbf{C}_d = \sigma^2 \mathbf{I}$ we get $t^* = \mathbf{1}^\top \mathbf{d} / \mathbf{1}^\top \mathbf{1} = \bar{d}$. $\square$

So the regularization path runs from the exact, noise-interpolating inverse at $\lambda = 0$ to a spatially uniform source field at $\lambda \rightarrow \infty$, whose level is the (precision-weighted) mean observation. Both endpoints are interpretable, which is a useful property when selecting λ from the trade-off curve between misfit $\|\mathbf{C}_d^{-1/2} (\mathbf{M}\hat{\mathbf{c}} - \mathbf{d})\|_2$ and roughness $\|\mathbf{P}\hat{\mathbf{c}}\|_2$ (Hansen 1998).

8 ERROR PROPAGATION

We now propagate the observational uncertainty $\mathbf{C}_d$ into the recovered sources. Because $\hat{\mathbf{c}}$ is an exactly linear function of $\mathbf{d}$, this is not a linearisation: it is exact.

Proposition 19 (Covariance of the sources). If $\mathbf{d}$ has covariance $\mathbf{C}_d$ and no constraint is active, then

$$\mathbf{C}_c = \mathbf{R}_\lambda \mathbf{C}_d \mathbf{R}_\lambda^\top. \quad (28)$$

Proof. For a deterministic matrix $\mathbf{R}$ and random vector $\mathbf{d}$, $\text{Cov}(\mathbf{R}\mathbf{d}) = \mathbf{R} \text{Cov}(\mathbf{d}) \mathbf{R}^\top$, directly from the definition of covariance (Menke 2012). Apply with $\mathbf{R} = \mathbf{R}_\lambda$, which is deterministic given the network and the fluxes. $\square$

Proposition 20 (Covariance of the modelled observations). The modelled observations $\hat{\mathbf{d}} = \mathbf{M}\hat{\mathbf{c}}$ have

$$\mathbf{C}_{\hat{\mathbf{d}}} = \mathbf{M} \mathbf{C}_c \mathbf{M}^\top = (\mathbf{M} \mathbf{R}_\lambda) \mathbf{C}_d (\mathbf{M} \mathbf{R}_\lambda)^\top, \quad (29)$$

and at $\lambda = 0$, $\mathbf{C}_{\hat{\mathbf{d}}} = \mathbf{C}_d$ exactly.

Proof. The first equality is Proposition 19 applied to the linear map $\hat{\mathbf{c}} \mapsto \mathbf{M}\hat{\mathbf{c}}$; the second follows by substitution. $\mathbf{M} \mathbf{R}_\lambda$ is the hat (influence) matrix. At $\lambda = 0$, $\mathbf{R}_0 = \mathbf{M}^{-1}$ so $\mathbf{M} \mathbf{R}_0 = \mathbf{I}$ and $\mathbf{C}_{\hat{\mathbf{d}}} = \mathbf{C}_d$. This is as it must be: by Proposition 9 the unregularized fit reproduces the observations exactly, so the modelled observations inherit their uncertainty unchanged. $\square$

8.1 The unregularized case is efficient

Proposition 21 (Cramér–Rao). At $\lambda = 0$,

$$\mathbf{C}_c = \mathbf{M}^{-1} \mathbf{C}_d \mathbf{M}^{-\top} = (\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M})^{-1}, \quad (30)$$

which is the inverse Fisher information of the model $\mathbf{d} \sim \mathcal{N}(\mathbf{M}\mathbf{c}, \mathbf{C}_d)$. The unregularized estimator therefore attains the Cramér–Rao lower bound.

Proof. For the identity, note $(\mathbf{M}^{-1} \mathbf{C}_d \mathbf{M}^{-\top})^{-1} = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M}$ by inverting each factor and reversing the order. For the Fisher information, the log-likelihood is $\ell(\mathbf{c}) = -\frac{1}{2}(\mathbf{d} - \mathbf{M}\mathbf{c})^\top \mathbf{C}_d^{-1} (\mathbf{d} - \mathbf{M}\mathbf{c}) + \text{const}$, so $\nabla_{\mathbf{c}} \ell = \mathbf{M}^\top \mathbf{C}_d^{-1} (\mathbf{d} - \mathbf{M}\mathbf{c})$ and $-\nabla_{\mathbf{c}}^2 \ell = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M}$, which is deterministic and hence equal to its expectation. The estimator is unbiased (see Proposition 22 with $\lambda = 0$), so the Cramér–Rao bound applies and is met with equality. $\square$

No estimator, weighted or otherwise, can do better than $\mathbf{M}^{-1}\mathbf{d}$ in the unbiased class. Section 4 already told us weighting cannot help; this tells us why nothing else can either. The only way to improve on (12) is to accept bias — which is exactly what regularization does.

8.2 Bias, resolution, and the two covariances

Proposition 22 (Bias and resolution). $\mathbb{E}[\hat{\mathbf{c}}] = \mathbf{R}_\lambda \mathbf{M} \mathbf{c}_{\text{true}}$, so the estimator has bias

$$\mathbf{b} = (\mathbf{R}_\lambda \mathbf{M} - \mathbf{I}) \mathbf{c}_{\text{true}} = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{c}_{\text{true}}, \quad (31)$$

and resolution matrix

$$\mathbf{R}_\lambda \mathbf{M} = \mathbf{I} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P}. \quad (32)$$

The mean-squared error is $\mathbb{E}[(\hat{\mathbf{c}} - \mathbf{c}_{\text{true}})(\cdot)^T] = \mathbf{C}_c + \mathbf{b}\mathbf{b}^T$.

Proof. Substituting $\mathbf{d} = \mathbf{M} \mathbf{c}_{\text{true}} + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ into (26) gives the first claim. Then

$$\mathbf{R}_\lambda \mathbf{M} - \mathbf{I} = \mathbf{A}_\lambda^{-1} \mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M} - \mathbf{I} = \mathbf{A}_\lambda^{-1} (\mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M} - \mathbf{A}_\lambda) = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P},$$

using (25). The MSE decomposition is the standard bias–variance identity (Aster et al. 2018; Hoerl and Kennard 1970). $\square$

The bias vanishes when $\lambda = 0$, and also whenever $\mathbf{P} \mathbf{c}_{\text{true}} = \mathbf{0}$, i.e. when the true source field really is uniform — the penalty is then telling the truth and costs nothing. Since $\mathbf{c}_{\text{true}}$ is unknown, $\mathbf{b}$ cannot be evaluated; the resolution matrix (32) is the standard computable proxy (Menke 2012), equal to $\mathbf{I}$ at $\lambda = 0$ and degrading as λ grows. Its trace, $\text{tr}(\mathbf{R}_\lambda \mathbf{M})$, is the effective number of degrees of freedom the data constrain.

This bias is precisely why (28) must be interpreted with care. It answers the question “if I repeated the measurements, how would my estimate scatter?” — and a heavily damped estimator scatters very little while being systematically wrong. A second covariance answers a different and often more useful question.

Proposition 23 (Posterior covariance). Read the penalty as a Gaussian prior with precision $\lambda \mathbf{P}$ and the likelihood as $\mathbf{d} \mid \mathbf{c} \sim \mathcal{N}(\mathbf{M} \mathbf{c}, \mathbf{C}_d)$. Then the posterior covariance is $\mathbf{A}_\lambda^{-1}$, and

$$\mathbf{R}_\lambda \mathbf{C}_d \mathbf{R}_\lambda^T = \mathbf{A}_\lambda^{-1} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{A}_\lambda^{-1}. \quad (33)$$

Consequently $\mathbf{C}_c \leq \mathbf{A}_\lambda^{-1}$, with equality if and only if $\lambda = 0$.

Proof. For the linear-Gaussian model the posterior precision is the sum of the likelihood precision $\mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M}$ and the prior precision $\lambda \mathbf{P}$, giving $\mathbf{A}_\lambda$ (Tarantola 2005). For the identity, substitute (26) and use $\mathbf{C}_d^{-1} \mathbf{C}_d \mathbf{C}_d^{-1} = \mathbf{C}_d^{-1}$:

$$\begin{aligned} \mathbf{R}_\lambda \mathbf{C}_d \mathbf{R}_\lambda^T &= \mathbf{A}_\lambda^{-1} \mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{C}_d \mathbf{C}_d^{-1} \mathbf{M} \mathbf{A}_\lambda^{-1} \\ &= \mathbf{A}_\lambda^{-1} (\mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M}) \mathbf{A}_\lambda^{-1} \\ &= \mathbf{A}_\lambda^{-1} (\mathbf{A}_\lambda - \lambda \mathbf{P}) \mathbf{A}_\lambda^{-1} = \mathbf{A}_\lambda^{-1} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{A}_\lambda^{-1}, \end{aligned}$$

using the symmetry of $\mathbf{A}_\lambda$. For the ordering, note that for any $\mathbf{x}$, $\mathbf{x}^T \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{A}_\lambda^{-1} \mathbf{x} = (\mathbf{A}_\lambda^{-1} \mathbf{x})^T \mathbf{P} (\mathbf{A}_\lambda^{-1} \mathbf{x}) \geq 0$ by Lemma 13, so the subtracted term is positive semi-definite. It is zero for all $\mathbf{x}$ only if $\lambda = 0$, since $\mathbf{A}_\lambda^{-1}$ is non-singular and $\mathbf{P} \neq \mathbf{0}$. $\square$

The propagated covariance is therefore always the smaller of the two, and quoting it alone at large λ understates the true error. The posterior covariance $\mathbf{A}_\lambda^{-1}$ accounts for what the damping costs as well as what it stabilises, and is the more honest error bar to report whenever $\lambda > 0$. At $\lambda = 0$ the two coincide and both equal the Cramér–Rao bound of Proposition 21.

9 SUMMARY

The practical picture is this. The mixing matrix is invertible, so the unmixing problem always has an exact solution — but that solution differences neighbouring observations, amplifying their errors by $\kappa_i = Q_i/q_i$, and interpolates the noise. Regularization trades this variance for bias, and because the penalty is a homogeneous quadratic the resulting estimator stays exactly linear in the data. That linearity is what delivers closed-form covariances, a resolution matrix, and effective degrees of freedom, none of which are available from the log-ratio formulation without Monte Carlo resampling. The price is the assumption of absolute rather than relative misfit — an assumption that Remark 12 shows is substantially repaired by weighting with a proportional error model, and which is in any case mild for tracers of low log-variance such as isotopic ratios (Blondes et al. 2016). For tracers whose information really is relative rather than absolute — that is, genuinely compositional data (Aitchison 1986) — the log-ratio formulation remains the better choice.

Two caveats bear repeating. Regularization is not optional: by Proposition 9 the unregularized problem reproduces the data exactly, noise included. And the analytical machinery of Section 8 is valid only while the constraint $\mathbf{c} \geq 0$ is inactive; when sites clamp, the estimator ceases to be linear and the reported covariances describe the unconstrained estimator instead.

Quantity	Expression
Mixing matrix	$M_{ij} = \Theta_{ij} \alpha_{j \rightarrow i} q_j / Q_i$
Exact inverse	$c_i = (Q_i d_i - \sum_{p \in A(i)} \alpha_{p \rightarrow i} Q_p d_p) / q_i$
Amplification	$\kappa_i = Q_i / q_i = 1 / M_{ii}$
Penalty operator	$\mathbf{P} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$
Normal matrix	$\mathbf{A}_\lambda = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} + \lambda \mathbf{P}$
Estimator	$\mathbf{R}_\lambda = \mathbf{A}_\lambda^{-1} \mathbf{M}^\top \mathbf{C}_d^{-1}$
Source covariance	$\mathbf{C}_c = \mathbf{R}_\lambda \mathbf{C}_d \mathbf{R}_\lambda^\top$
Prediction covariance	$\mathbf{C}_d = \mathbf{M} \mathbf{C}_c \mathbf{M}^\top$
Posterior covariance	$\mathbf{A}_\lambda^{-1}$
Resolution	$\mathbf{R}_\lambda \mathbf{M} = \mathbf{I} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P}$
Bias	$\mathbf{b} = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{c}_{\text{true}}$

Table 1: **Principal results.** At $\lambda = 0$: $\mathbf{R}_0 = \mathbf{M}^{-1}$, the resolution matrix is $\mathbf{I}$, the bias vanishes, the two covariances coincide, and $\mathbf{C}_c$ attains the Cramér–Rao bound.

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